Prediction Market Contracts Are Bonds

A Reduced-Form Stochastic Intensity Model

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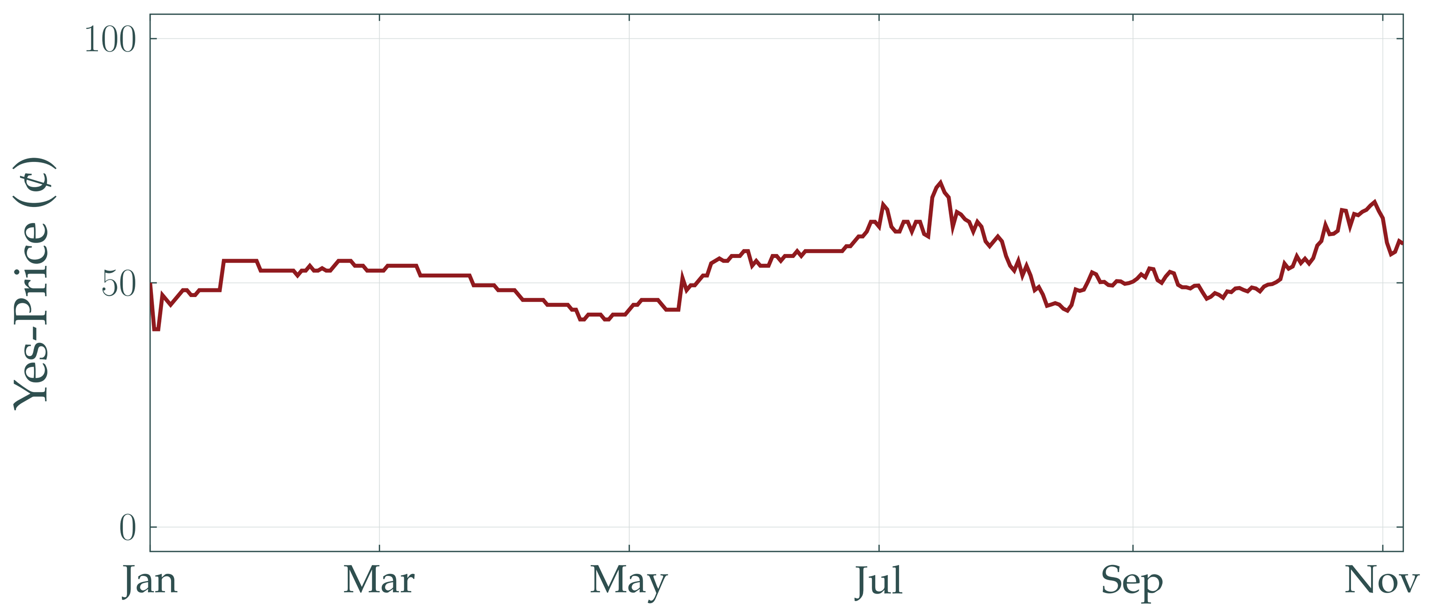
Contributions. I construct a novel pricing framework for timing prediction markets by showing an equivalence with bond pricing. A latent jump-diffusion intensity process yields a tractable pricing equation, with a Kalman filter-based parameter estimation procedure. I show fair calibration on four real markets, even when point prediction is difficult.

Timing Markets Need Their Own Model

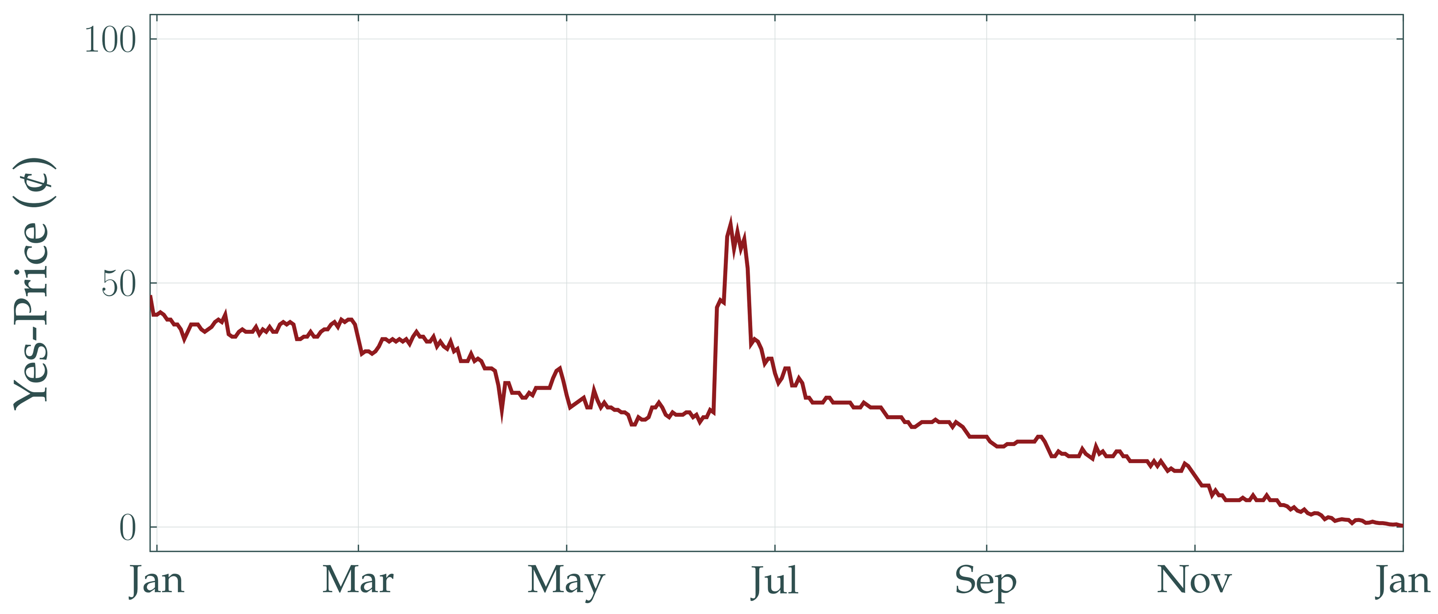
Timing prediction markets are based on questions like "Will event A occur before time T?" The market's yes-price π_t^{yes} and no-price π_t^{no} are the prices of contracts paying out \$1 if the answer is yes and no, respectively. We interpret market yes-prices as estimates of the probability of A occurring before T.

Unlike *election-style* markets, based on questions like "Will outcome A or B occur?", where prices stay uncertain until resolution, in timing markets the yes-price $\pi_t^{\text{yes}} \to 0$ as $t \to T$, mechanically.

Will Trump Win the 2024 U.S. Presidential Election?



Khamenei Out as Supreme Leader of Iran in 2025?



Unlike election-style markets, timing market prices decay to zero near the deadline. No existing model captures this convergence behavior.

The Pricing Equation

At time t consider a timing market with end date T. Suppose that the underlying event A arrives at time R with an inhomogeneous Poisson process N_t having intensity λ_t. The no-price compatible with interpreting prices as probabilities is then

$$\pi_t^{\text{no}} = \mathbb{P}\left[R > T \middle| \mathcal{F}_t\right] = \mathbb{E}\left[e^{-\int_t^T \lambda_s \mathrm{d}s} \middle| \mathcal{F}_t\right].$$

This is the same pricing equation as for reduced-form models of defaultable corporate bonds and for zero-coupon bonds with stochastic short rate.

Modeling the Intensity

Specifying the intensity dynamics is the key modeling decision. I use a Cox-Ingersoll-Ross process with compound Poisson jumps:

$$\mathrm{d}\lambda_t = \kappa(\mu - \lambda_t)\mathrm{d}t + \sigma\sqrt{\lambda_t}\mathrm{d}W_t + \mathrm{d}J_t,$$

where $J_t = \sum_{k=1}^{M_t} Y_k$, M_t is a Poisson process with mean α, and $Y_k \sim \mathrm{Exp}(\gamma)$ i.i.d. Strictly positive jumps are unrealistic but guarantee that $\lambda_t \geq 0$. The model suffers from weak identifiability of σ, κ, α and γ.

Because the JCIR-process is affine we get a closed form solution to the pricing equation

$$\pi_t^{\text{no}} = \mathbb{E}\left[e^{-\int_t^T \lambda_s \mathrm{d}s} \middle| \mathcal{F}_t\right] = A_t e^{-B_t \lambda_t},$$

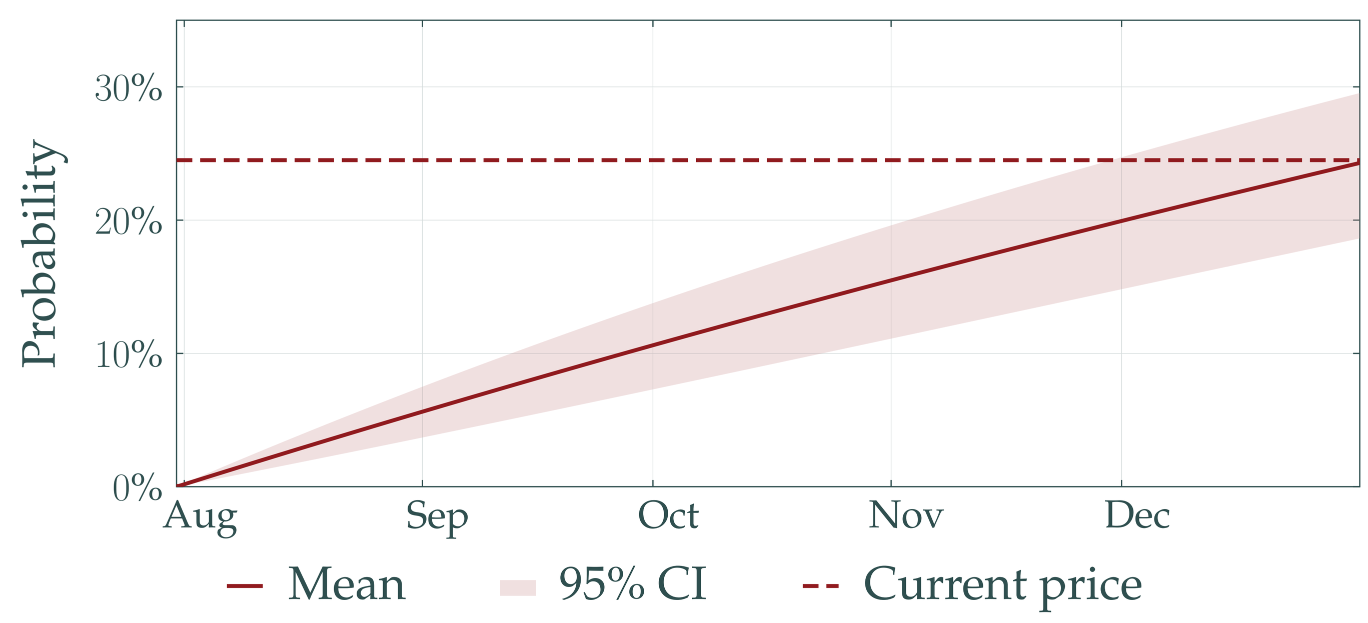
for A_t, B_t which depend on the model parameters and the end date T.

Parameters are found by constructing a linear Gaussian approximation of this state space system and running a Kalman filter on transformed price data; the likelihood maximizing the quasi-likelihood of the filter residuals yields estimates.

When Will It Resolve?

The probability of A having occurred by any date s is simply the price of a yes-contract with end date s.

Probability of Resolving by Date



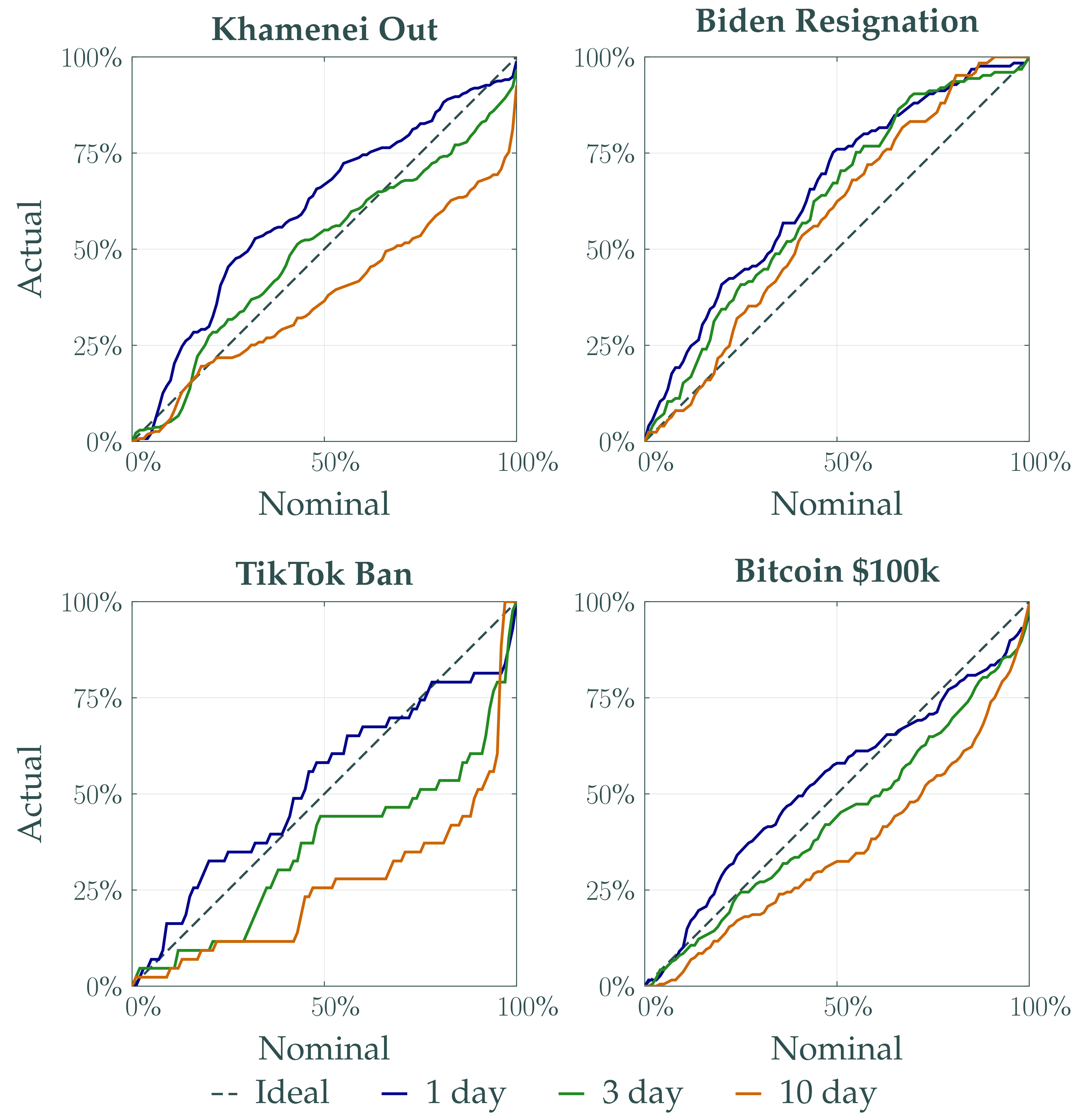
These resolution probabilities are not available without a pricing model.

Does It Work?

The model is evaluated on four real timing markets, regarding the ousting of Khamenei, the resignation of President Biden, the banning of TikTok in the U.S., and the price of Bitcoin reaching \$100,000. Parameters are re-estimated daily on expanding windows starting with 50 days of data. Weak identification forces us to hold some parameters fixed.

Calibration across the entire predictive distribution can be inferred from coverage rates of prediction intervals at 1, 3, and 10-day horizons.

Coverage Rates of Prediction Intervals



Across all four markets JCIR point forecasts struggle to beat naive persistence, because log-returns exhibit little autocorrelation. The model's value lies in calibrated predictive distributions, not point accuracy.

Predictive distributions achieve reasonable calibration on multiple markets.

Conclusion

The framework accommodates alternative intensity specifications—e.g. mean-reverting levels or jumps of separate sizes for distinct news types—and stabilizing estimation via parametric bootstrapping or non-linear filtering. Joint estimation across markets on the same event would improve identification and expose inter-market pricing inconsistencies.